



Comprehension of Technical Materials

Outcomes of the class

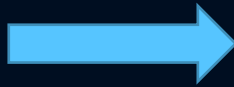
- Learn about the difference between reading general material and technical material.
- Learn about important techniques for reading, interpreting as well as writing technical content.

Difference between general and technical material

- Reading technical material is different from reading general material.
- Example: *You can begin reading from page 3 in a newspaper and still understand what is being discussed.*
- You cannot follow the same strategy with technical material .
- Moreover, the **audience** and the **purpose** is different in both the cases.

Reading techniques which are common for both kinds of reading

STEP 1



STEP 2

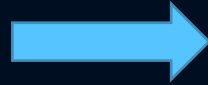
- Prepare for reading.
 - HOW?
 - Predict the information you are going to receive.
 - Recall the information you know about the topic.
- Scan and skim the text
 - Scan: look at graphic information, look at the bibliography, look at the notes/abstract.
 - Skim: Look at the different paragraphs and see if you can identify the topic sentence.

Activity 1

- Compare a news article vs. a technical abstract.
- What differences do you notice in writing style?

Techniques specific to technical material

STEP 3



STEP 4

- Read the text thoroughly.
 - **Tip:** consult a glossary if there are any unfamiliar terms.
 - Re-read if you are unable to understand a sentence.
- Make notes.
 - **Tip:** Make notes of two articles. Use different strategies in both. See which one saves you time.

STEP 5

- Review what you have read
- See you are able to recall what you have just read.

Q 1. Can you recall some information in a book or article you read recently?

Q 2. Why were you able to recall it?

Two useful reading methods

- ERRO reading technique.
Developed in 1985.
- It has four stages;
Estimate, Read ,
Respond , Question.

Stage	Description
1) Estimate	Estimate what the text will be like. Rapid reading techniques may be used for this purpose.
2) Read	Read the text carefully and thoroughly
3) Respond	Respond to the text
4) Question	Question things about the text and analyses responses to it.

SQ₃R reading technique

Stage	Description
1) Survey	Glancing rapidly through the text before reading any part of it.
2) Question	Asking appropriate questions for each part of the text in order to give a purpose to the reading.
3) Read	Read carefully and make notes.
4) Recall	Recall the content at the end of each part and amend notes.
5) Revise	Checking the accuracy of reading, recall and reviewing notes.

The purpose of different graphic aids for communicating technical information

Graphic aids	Function
Tables	Organize data into groups and show quantitative data and related information
Bar graphs	Show comparative or contrastive information
Tree diagrams	Show classificatory data
Flow diagrams*	Show the steps of a process
Pie charts	Show information related to proportion or percentages.
Flow charts*	Present complex information

*The purpose of flow charts and flow diagrams might be similar in some cases

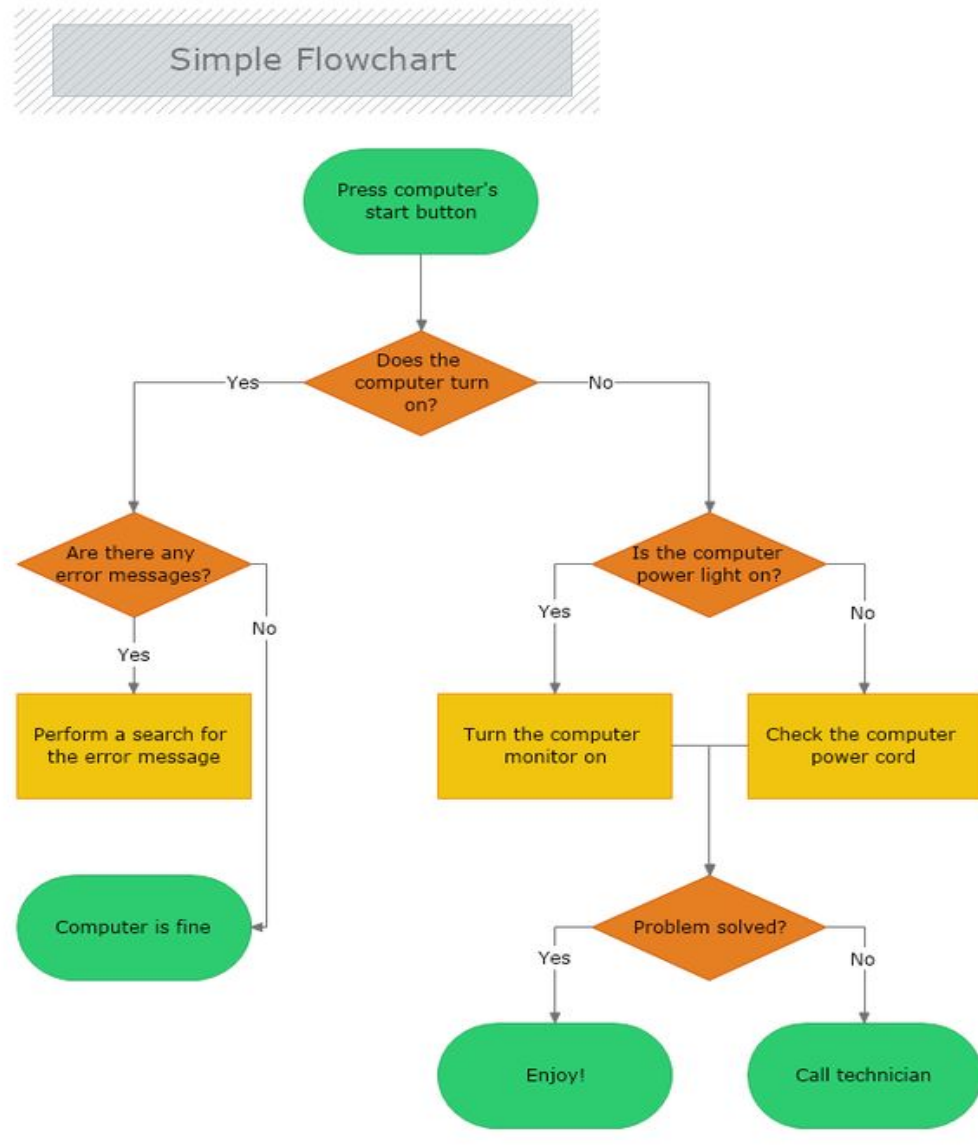


Fig.1 Flow chart

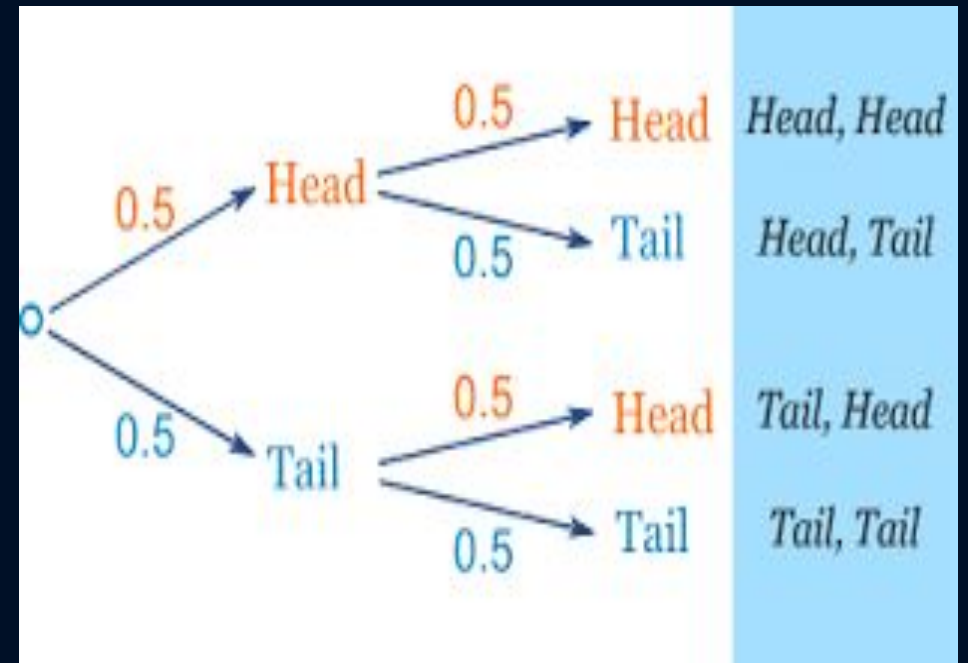


Fig. 2 Tree diagram showing probability

Activity 2: INTERPRETING A TABLE

Components of a Microscope		Functions of each component
Optical parts	Eyepiece Body tube Objectives	Serves as a magnifying glass and limits the field Refracts the light Magnifies the image
Mechanical Parts	Coarse and fine adjustment dials Nose-piece Metal plate	Provide clear image Holds the objectives The specimen is placed on it
Illuminating parts	Illumination source Condenser	Provides necessary light Intensifies the light

Read the following statements and mark True or False in the light of the above table.

- (a) A microscope is an optical instrument for magnifying micro-organisms.
- (b) A microscope does not consist of any mechanical part.
- (c) The body tube serves as a magnifying glass and limits the field seen by the eye.
- (d) The condenser provides the necessary light.
- (e) The coarse and fine adjustment dials provide a clear image.
- (f) The eyepiece refracts light.
- (g) The nose-piece holds the objective.
- (h) We place the specimen on a metal plate.
- (i) The objective magnifies the image.
- (j) The illumination source intensifies the light

Answers for activity given on slide 12

- ☐ A) true
- ☐ B) false
- ☐ C) False
- ☐ D) False
- ☐ E) True
- ☐ F) False
- ☐ G) True
- ☐ H) True
- ☐ I) True
- ☐ J) False

Reading Instructions and Technical Manuals

- Where have you read instructions/ instructional manuals?
- Have you read one from cover to cover?
- Why do you skip some of the pages in a manual while reading?

Reading technical manuals

Here are a few key steps to remember when reading manuals.

- Scan the graphic information.
- While reading the text, look for safety instructions. Identify the precautions.
- Consult a glossary if it is available. If not, then circle or underline the scientific/ technical terms you are unfamiliar with.
- Look for contextual signals that might help you identify the meaning of the word.

Answers for class activity (UPS)

1. a) True

- b) False
- c) False
- d) True
- e) False
- f) True

2 a) iii, b) i, c) ii, d) iii c) ii

Answers for class activity

3. a) In colder territories allow the UPS to come to room temperature before operating.

B) The UPS charges its battery whenever it is connected to utility power.

C) If the UPS is used without charging the battery, run time may be reduced.

D) A laser printer periodically draws significantly more power when idle, and will overload the UPS.

E) With the UPS unplugged from the wall outlet, press the OFF button for two seconds in order to reset the internal microprocessor

F) The UPS is operating whenever it is plugged in and mains are on.

G) the self-test can be used to verify, both, the operation of the UPS and the condition of the battery.

H) The UPS automatically corrects high and low utility voltages so that the loads receive voltage within the normal range.

I) When the UPS is off and there is no utility power, the cold start is used to apply power to the loads from the UPS's battery.